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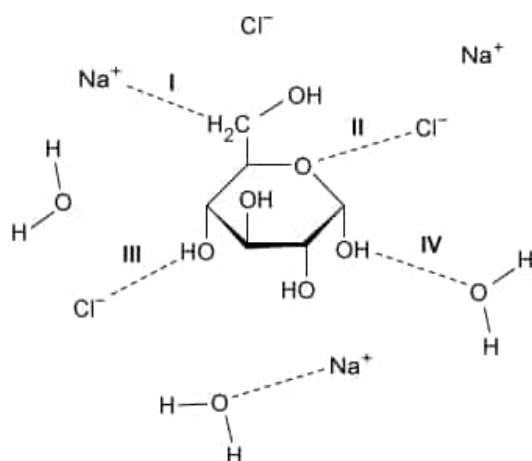
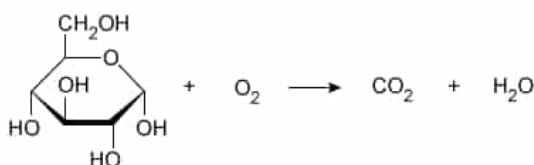


Figure 1 Diagram of selected intermolecular interactions in a saline solution containing dextrose ($C_6H_{12}O_6$)

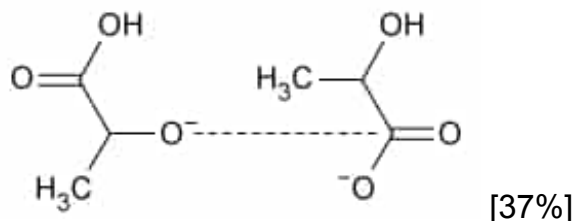
Dextrose is metabolized by a multistep process that results in the overall chemical conversion shown by the unbalanced equation in Reaction 1.



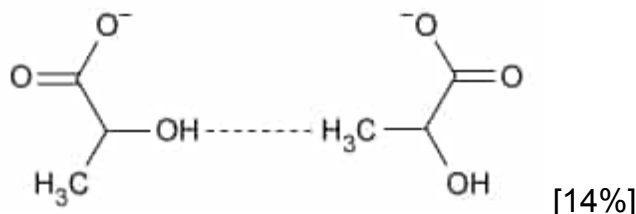
Reaction 1

Which of the following pairs of Lewis structures best represents a dipole-dipole interaction between permanent dipoles in two lactate molecules in a Plasma-Lyte A solution?

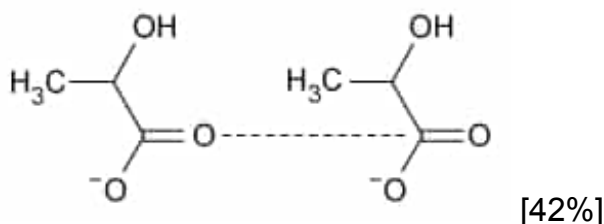
☐ A.



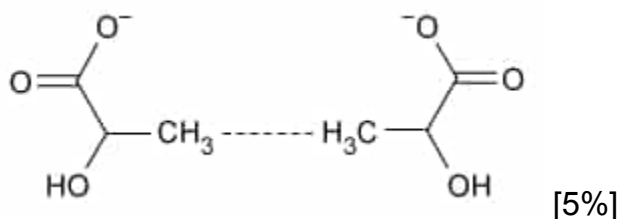
☐ B.



☒ C.



☐ D.

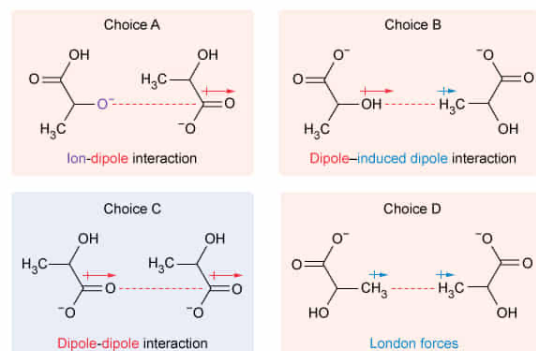


Correct

43% answered correctly

Explanation:

Diagrams of noncovalent intermolecular interactions



A **dipole** consists of opposite electric charges separated across a distance (eg, a chemical bond). In molecules, **permanent dipoles** form across some **covalent bonds** due to **unequal sharing of electrons** between two atoms caused by large differences in **electronegativity**. In a dipole, the region with greater electron density gains a partial negative charge (δ^-) and the region with less electron density acquires a partial positive charge (δ^+).

Many **intermolecular forces** (ie, noncovalent interactions between atoms and molecules) involve the partial charges of bond dipoles. A **dipole-dipole interaction** occurs when the opposite partial charges of the permanent dipoles of the polar bonds in neighboring molecules form a mutual attraction. In the structure of lactate, the C=O bond is polar and exhibits a permanent dipole. As such, the opposite partial charges of two C=O bonds can align and participate in dipole-dipole interactions, as shown in **Choice C**.

(Choice A) This pair of structures represents an **ion-dipole interaction** in which the full charge of an ion is attracted to the opposite partial charge of a polar bond dipole. In addition, the incorrect oxygen atom is deprotonated in one of the lactate structures.

(Choice B) This pair of structures represents a **dipole-induced dipole interaction** in which the permanent dipole from the polar O–H bond distorts the electron cloud of a neighboring nonpolar C–H bond and induces a weak temporary dipole. Thus, the O–H dipole and the induced C–H dipole form a weak attraction.

(Choice D) This pair of structures represents **London forces** that form when two nonpolar bonds induce *momentary dipoles* in each other by distortions in the electron cloud around the atoms. These short-lived instantaneous dipoles invoke weak mutual attractions.

Educational objective:

Ion-dipole interactions occur between the partial charge of a permanent dipole and the full charge of an ion. Dipole-induced dipole interactions occur when a permanent dipole induces a weak temporary dipole in a nonpolar bond or atom. Dipole-dipole interactions occur between the partial charges of two permanent dipoles.

Time spent:0 sec

Subject:
General Chemistry

QID:403135

System:
Interactions of Chemical Substances

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Lactate is included in some solutions, such as Plasma-Lyte A, to enhance electrolyte balance. Lactate is metabolized into bicarbonate within the liver, which can be advantageous to counteract acidic conditions in the body.

Dextrose solutions may be administered to provide a source of carbohydrates that can sustain blood glucose and reduce the depletion of liver glycogen during fluid maintenance. A diagram representing a dextrose saline solution and several possible intermolecular interactions is shown in Figure 1.

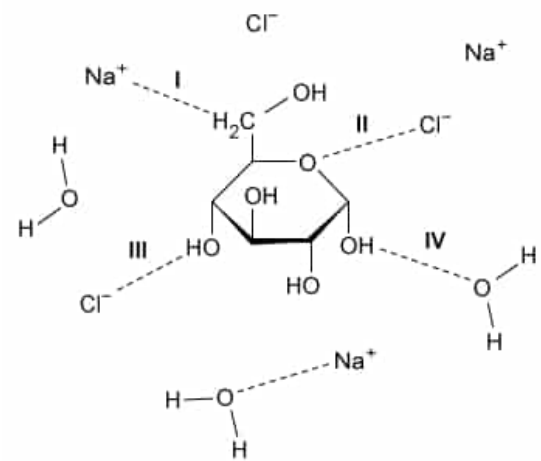
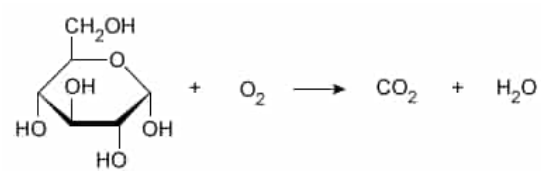


Figure 1 Diagram of selected intermolecular interactions in a saline solution containing dextrose (C₆H₁₂O₆)

Dextrose is metabolized by a multistep process that results in the overall chemical conversion shown by the unbalanced equation in Reaction 1.



Reaction 1

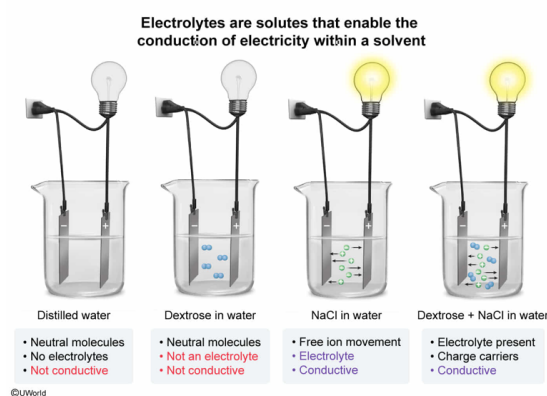
Electrolytes are solutes that enable the conduction of electricity within a solvent. Based on its composition, the dextrose saline solution from Table 1 is:

- ☐ A. conductive, because dextrose is an electrolyte. [3%]
- ✓ ☒ B. conductive, because sodium chloride is an electrolyte. [82%]
- ☐ C. conductive, because both dextrose and sodium chloride are electrolytes. [11%]
- ☐ D. nonconductive, because neither dextrose nor sodium chloride are electrolytes. [2%]

Correct

81% answered correctly

Explanation:



Chemical compounds can be **classified** by their bonding characteristics and physical properties:

- **Ionic compounds** consist of oppositely charged **metal and nonmetal** ions bound by strong **electrostatic attractions** (ie, **ionic bonds**) that result in solids with very high melting points. Ionic compounds do not conduct electricity as solids because the **ions are held in place**, but as liquids (ie, when melted or dissolved in solution) ionic compounds are conductive because the ions can **move freely** and act as charge-carrying **electrolytes**.
- **Metallic compounds** consist of metal nuclei surrounded by a delocalized "**sea**" of **electrons** that can move freely. These strong metallic bonds give metals a high melting point and make them **electrically conductive** in both solid and liquid states.
- **Molecular compounds** consist of **covalently bonded** nonmetals that have weak, noncovalent, **intermolecular interactions** between neighboring molecules that result in relatively low melting points (less than ~300 °C). Because the bonding electrons are **localized** (ie, confined within the covalent bonds), molecular compounds are generally nonconductive as both solids and liquids.

Dextrose is not an electrolyte and does not conduct electricity in solution because it is a **molecular compound** made of covalently bonded nonmetal atoms that do not form ions in solution. However, **sodium chloride is an electrolyte** because it is an **ionic compound** made of metal and nonmetal ions that dissociate in solution. Therefore, the presence of the sodium chloride electrolyte causes the dextrose saline solution to be electrically conductive.

(Choices A, C, and D) The dextrose saline solution is a mixture of an electrolyte (sodium

chloride) and a nonelectrolyte (dextrose). Although dextrose does not support conductivity, the presence of sodium chloride makes the solution electrically conductive.

Educational objective:

Soluble ionic compounds act as electrically conductive electrolytes in solution, but covalent molecular compounds generally do not conduct electricity in solution.

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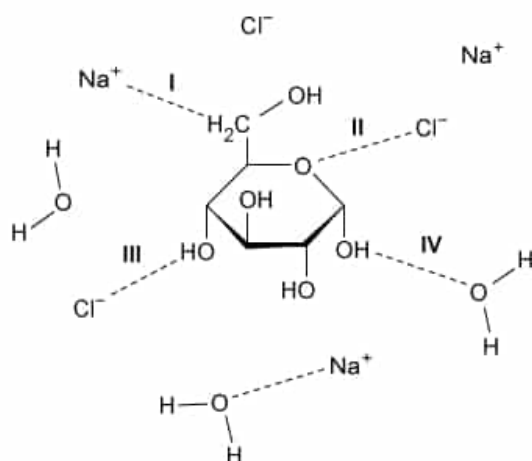
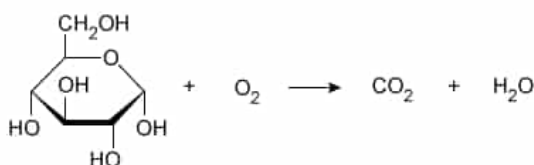


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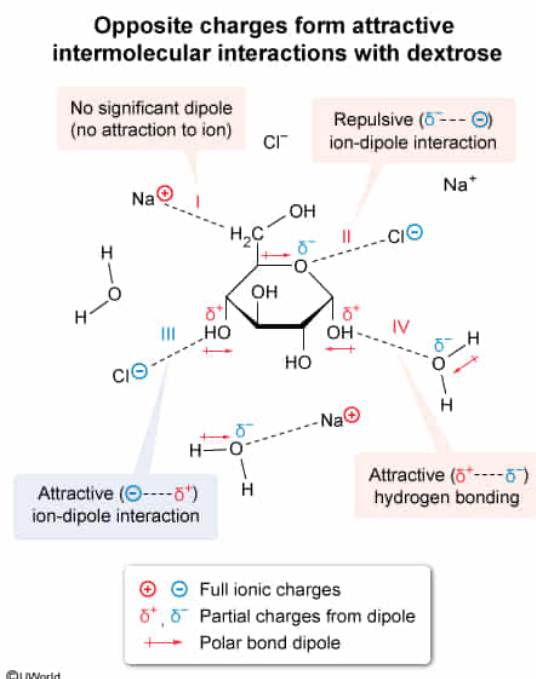
Which labeled intermolecular interaction in Figure 1 represents an attractive ion-dipole interaction?

- ☐ A. I [12%]
☐ B. II [24%]
✓ ☒ C. III [58%]
☐ D. IV [4%]

Correct

58% answered correctly

Explanation:



Permanent **dipoles** (ie, opposite electric charges separated across a distance) form across **covalent bonds** that have an **unequal sharing of electrons** between two atoms due to a significant **difference in electronegativity**. The more electronegative atom has more electron density and a partial negative charge (δ^-), and the less electronegative atom has less electron density and a partial positive charge (δ^+).

Many **intermolecular forces** (ie, noncovalent interactions between atoms and molecules) involve attractions between the partial charges of bond dipoles. **Opposite charges attract** whereas like charges repel; as such, charged ions can form **ion-dipole interactions** in which the full charge of the ion is attracted to the opposite partial charge (and repelled by the like partial charge) of a polar bond dipole.

In the structure of dextrose, the C–O and O–H bonds are polar (ie have a permanent dipole) due to the large difference in **electronegativity** between the atoms. As such, only interactions II and III are ion-dipole interactions. Hydrogen is less electronegative than oxygen, so hydrogen has less electron density and carries a partial positive charge.

Therefore, interaction III is attractive because the partial positive charge of the dipole (on the hydrogen atom) is oriented toward a negatively charged chloride ion (ie, opposite charges attract).

(Choice A) Interaction I cannot be a typical ion-dipole interaction because C–H bonds are nonpolar (ie no significant permanent dipole).

(Choice B) Interaction II is a *repulsive* ion-dipole interaction because the partial negative charge on the oxygen atom is oriented toward a negatively charged chloride ion (ie, like charges repel).

(Choice D) Interaction IV is attractive but results from a special type of dipole-dipole interaction called **hydrogen bonding**. Interaction IV cannot be an ion-dipole interaction because it does not involve an ion (ie, no fully charged species).

Educational objective:

Ion-dipole interactions occur between the partial charge of a permanent dipole and the full charge of an ion. Opposite charges attract whereas like charges repel.

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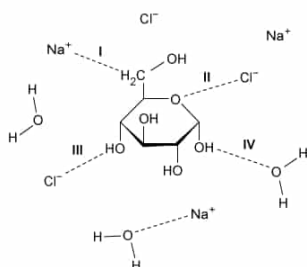
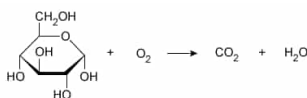


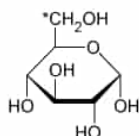
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Reaction 1

During an investigation of metabolic rate, isotopically labeled dextrose (shown below) was prepared with a carbon-13 atom in the substituent (indicated by *) to function as a radioactive tracer.



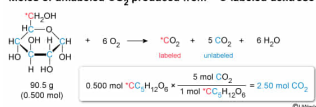
If 90.5 g of isotopically labeled dextrose is fully metabolized into carbon dioxide and water, how many moles of unlabeled carbon dioxide are produced? (Note: The molar mass of the ^{13}C -labeled dextrose is 181 g/mol.)

- ☐ A. 0.42 mol [9%]
☐ B. 0.50 mol [34%]
☒ C. 2.50 mol [28%]
☐ D. 3.00 mol [26%]

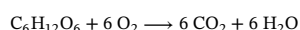
Correct
 29% answered correctly

Explanation:

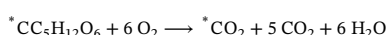
Moles of unlabeled CO_2 produced from ^{13}C -labeled dextrose



For any chemical reaction, the **law of conservation of mass** states that atoms are neither created nor destroyed; they are only rearranged. Consequently, the **same number** of each type of atom must be present **on both sides** of a reaction arrow (ie, the reaction must be **balanced**). In the passage, Reaction 1 is unbalanced and provides only a summary of the species involved in the metabolic process. Balancing Reaction 1 gives:



If the **isotopically labeled** dextrose specified in this question is used in the reaction, then 1 out of every 6 carbon atoms is a ^{13}C atom (indicated by *). Accordingly, only 1 in 6 CO_2 molecules produced will be isotopically labeled with a ^{13}C atom (ie, 5 out of 6 CO_2 molecules will be **unlabeled**).



The question states 90.5 g of labeled dextrose reacts. Dividing by the **molar mass** (ie, the **mass of 1 mole** of the compound) gives the number of moles that react:

$$90.5 \text{ g } ^*\text{C}_5\text{H}_{12}\text{O}_6 \times \frac{1 \text{ mol } ^*\text{C}_5\text{H}_{12}\text{O}_6}{181 \text{ g } ^*\text{C}_5\text{H}_{12}\text{O}_6} = 0.500 \text{ mol } ^*\text{C}_5\text{H}_{12}\text{O}_6$$

The amounts of reactants and products in a reaction are related to one another by the **mole ratios** from a balanced reaction equation. Multiplying the moles of labeled dextrose by the mole ratio from the balanced isotopically labeled reaction gives the moles of unlabeled CO₂ produced.

$$0.500 \text{ mol } ^*\text{C}_5\text{H}_{12}\text{O}_6 \times \frac{5 \text{ mol CO}_2}{1 \text{ mol } ^*\text{C}_5\text{H}_{12}\text{O}_6} = 2.50 \text{ mol CO}_2$$

Alternatively, the moles of unlabeled CO₂ can be calculated by first determining the total moles of CO₂ produced (ie, labeled + unlabeled) and then using the ratio of unlabeled CO₂ to total CO₂.

$$0.500 \text{ mol } ^*\text{C}_5\text{H}_{12}\text{O}_6 \times \frac{6 \text{ mol } (^*\text{CO}_2 + \text{CO}_2)}{1 \text{ mol } ^*\text{C}_5\text{H}_{12}\text{O}_6} = 3.00 \text{ mol } (^*\text{CO}_2 + \text{CO}_2)$$

$$3.00 \text{ mol } (^*\text{CO}_2 + \text{CO}_2) \times \frac{5 \text{ mol CO}_2}{6 \text{ mol } (^*\text{CO}_2 + \text{CO}_2)} = 2.50 \text{ mol CO}_2$$

(Choice A) 0.42 mol of unlabeled CO₂ is the number calculated using the 1:1 mole ratio of total CO₂ to dextrose from the *unbalanced* version of Reaction 1 and then applying the correct 5:6 mole ratio of unlabeled CO₂ to total CO₂.

(Choice B) 0.50 mol is the number of moles of *labeled* *CO₂ produced in the reaction.

(Choice D) 3.00 mol is the total number of moles of carbon dioxide produced (ie, *CO₂ + CO₂).

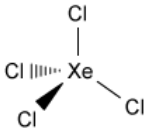
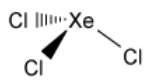
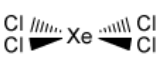
Educational objective:

The molar mass permits conversions between a given amount of mass and the number of moles contained in that mass. Mole ratios from a balanced reaction equation can be applied to determine the number of moles of one chemical species relative to the moles of another chemical species in a reaction.

Time spent: 0 sec
Subject:
 General Chemistry

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System:
 Interactions of Chemical Substances

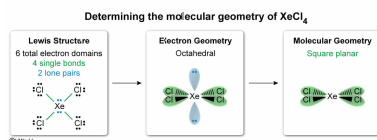
Considering all nonbonding and bonding electrons, which structure represents the molecular geometry of XeCl_4 ?

- ☐ A.
- 
- Tetrahedral [44%]
- ☐ B.
- 
- Trigonal pyramidal [1%]
- ☐ C.
- 
- See-saw [11%]
- ☒ D.
- 
- Square planar [42%]

Correct

42% answered correctly

Explanation:



Valence-shell electron-pair repulsion (VSEPR) theory states that the spatial arrangement of atoms in a molecule is determined by maximizing the separation between the electron domains (**bonding** and **nonbonding** pairs of **valence electrons**) around the central atom to minimize the repulsive forces.

The **electron geometry** of a molecule describes the arrangement of electron domains whereas the **molecular geometry** describes the arrangement of bonded atoms. The molecular geometry depends on the number of bonding versus nonbonding (lone pair) electron domains. Lone pairs exert more repulsion than bonding electrons, resulting in a compression of bond angles relative to the electron geometry. As such, if two molecules have the same number of electron domains but differ in the number of lone pairs, they will have different molecular geometries.

In this question, the molecule XeCl_4 has a total of 36 valence electrons. The central Xe atom is an exception to the **octet rule** and can employ an **expanded octet** by filling empty *d*-orbitals. Consequently, the **Lewis structure** for XeCl_4 shows that Xe has two pairs of nonbonding electrons and four Xe–Cl bonds (ie, a total of six electron domains). These six electron domains give XeCl_4 an **octahedral** electron geometry but because two of the six electron domains are nonbonding, the molecular geometry of XeCl_4 is **square planar**.

(Choice A) A **tetrahedral** molecular geometry results from four bonds and *zero* nonbonding electrons. XeCl_4 has *two pairs* of nonbonding electrons.

(Choice B) A **trigonal pyramidal** molecular geometry results from *three* bonds and *one pair* of nonbonding electrons. XeCl_4 has *four* bonds and *two pairs* of nonbonding electrons.

(Choice C) A **see-saw** molecular geometry results from four bonds and *one pair* of nonbonding electrons.

Educational objective:

The valence-shell electron-pair repulsion (VSEPR) model can be used to predict molecular geometry. A square planar geometry forms when four bonding and two nonbonding electron domains are oriented around the central atom of a molecule.

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